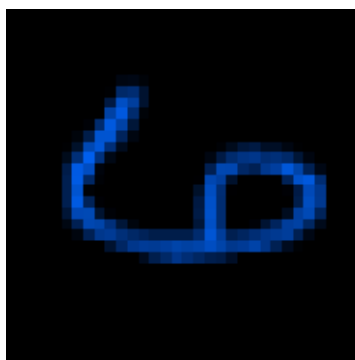
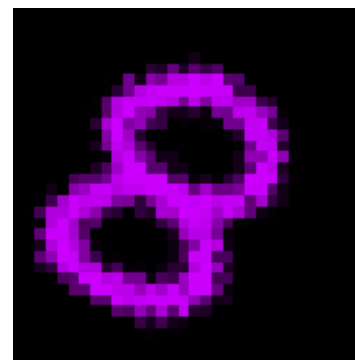
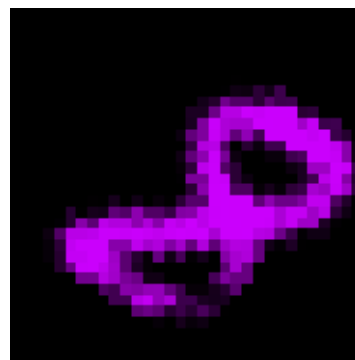
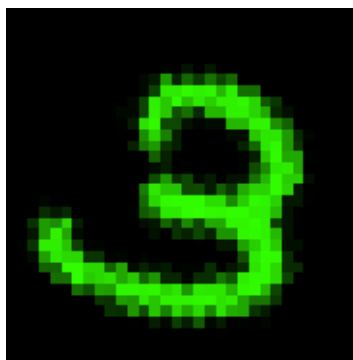
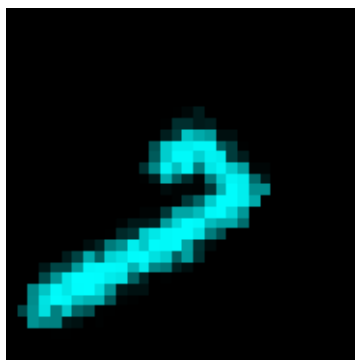
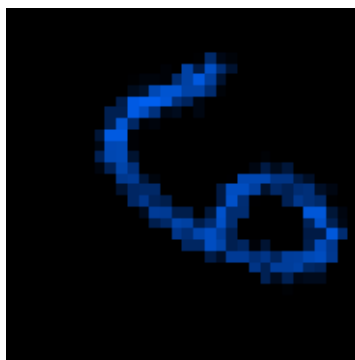


CMNIST Shift Visuals

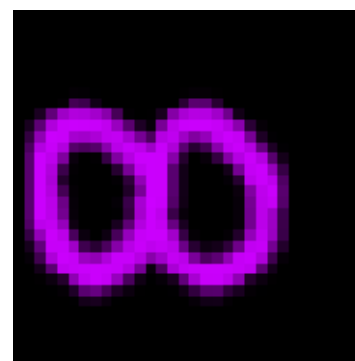
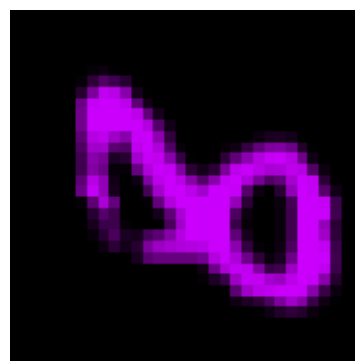
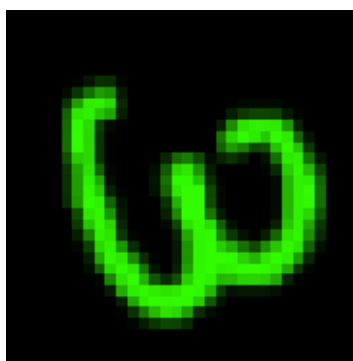
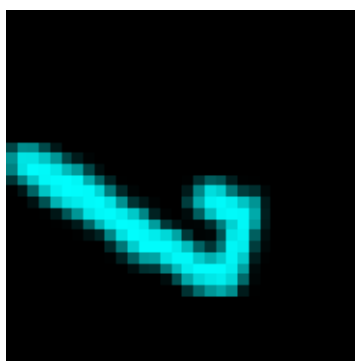
Original
(Huber 0)



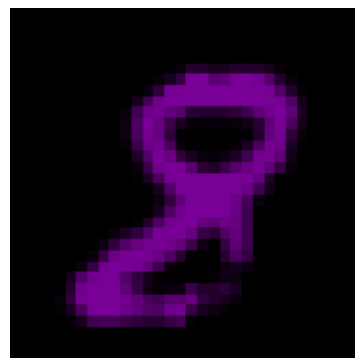
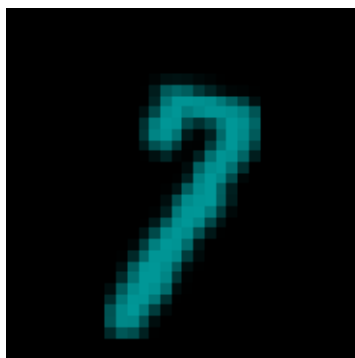
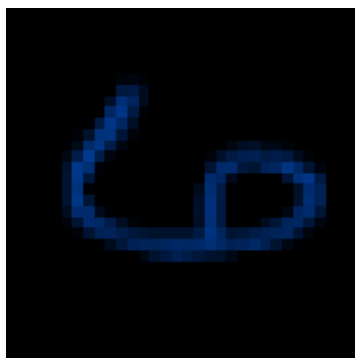
Rotation (LOW)
30.0 deg



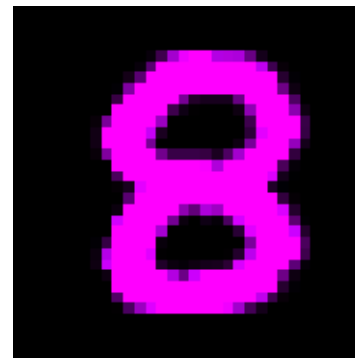
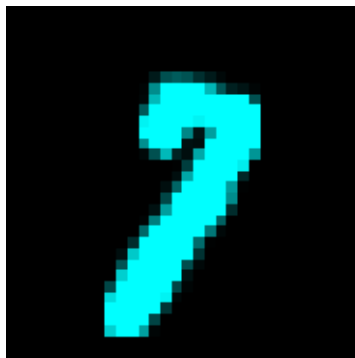
Rotation (HIGH)
90.0 deg



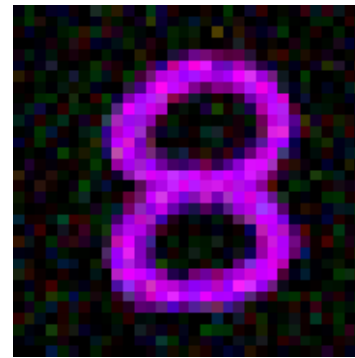
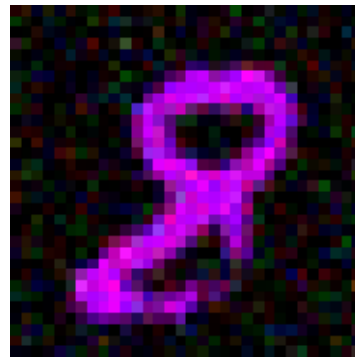
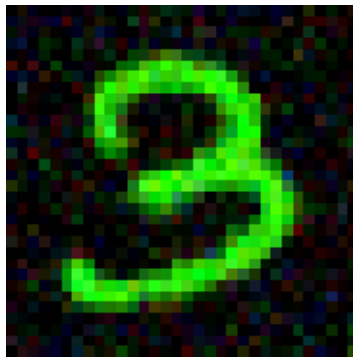
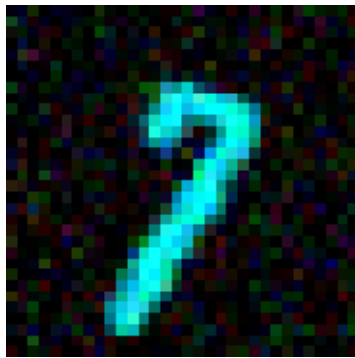
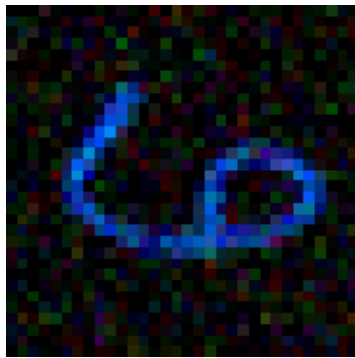
Lighting (LOW)
Bright=0.6



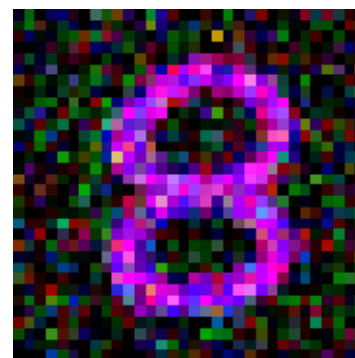
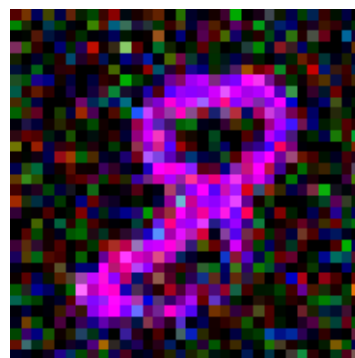
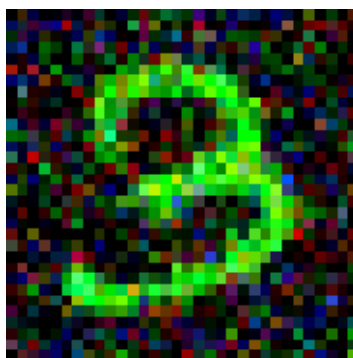
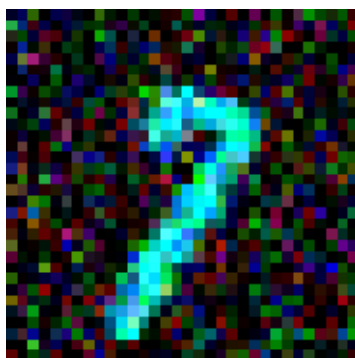
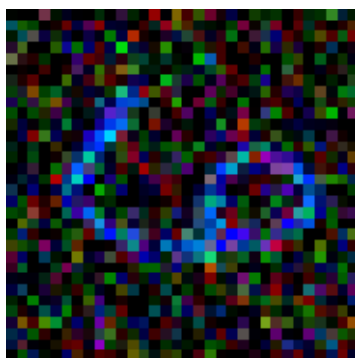
Lighting (HIGH)
Bright=3.0



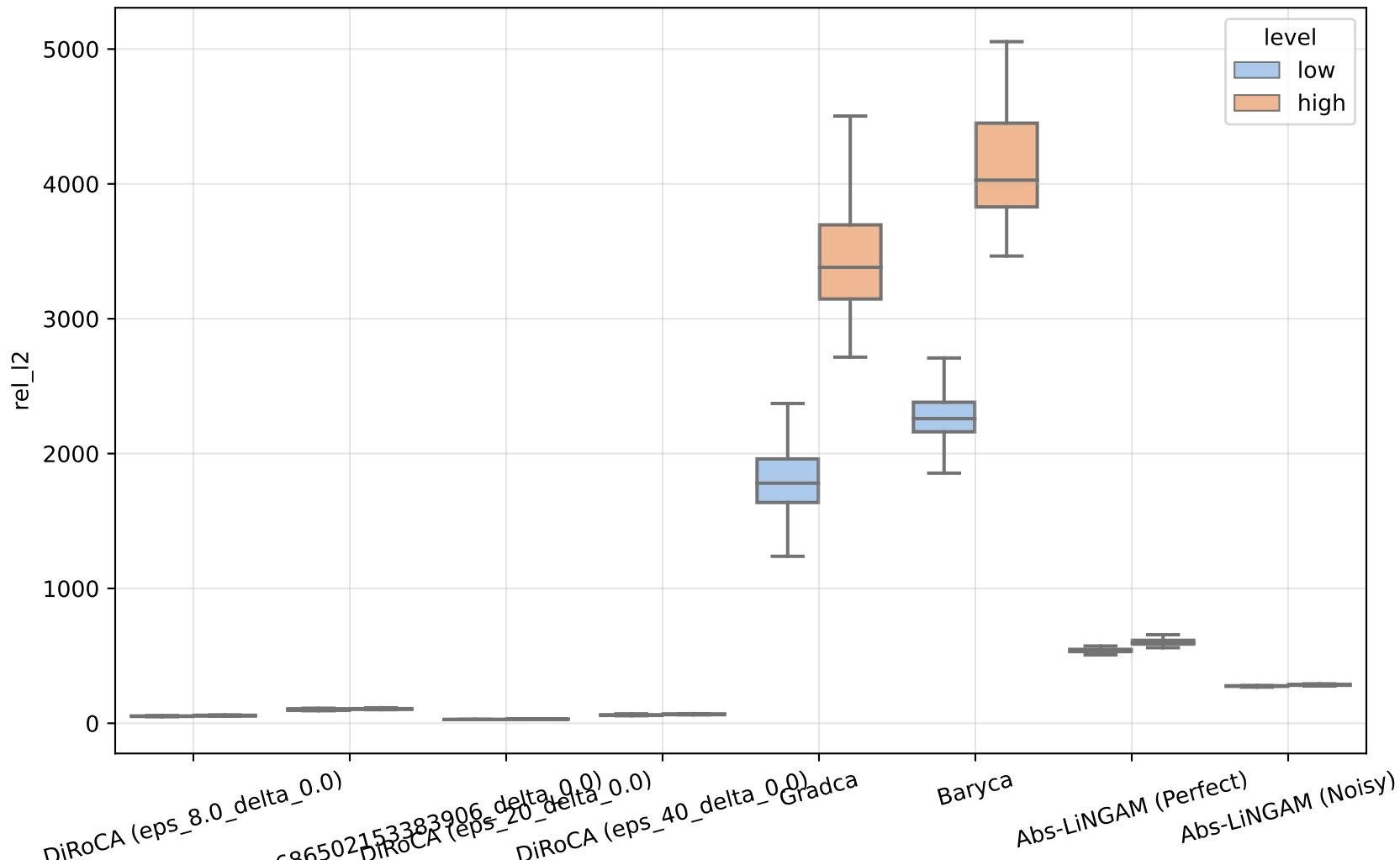
Huber
($\sigma=0.2$)



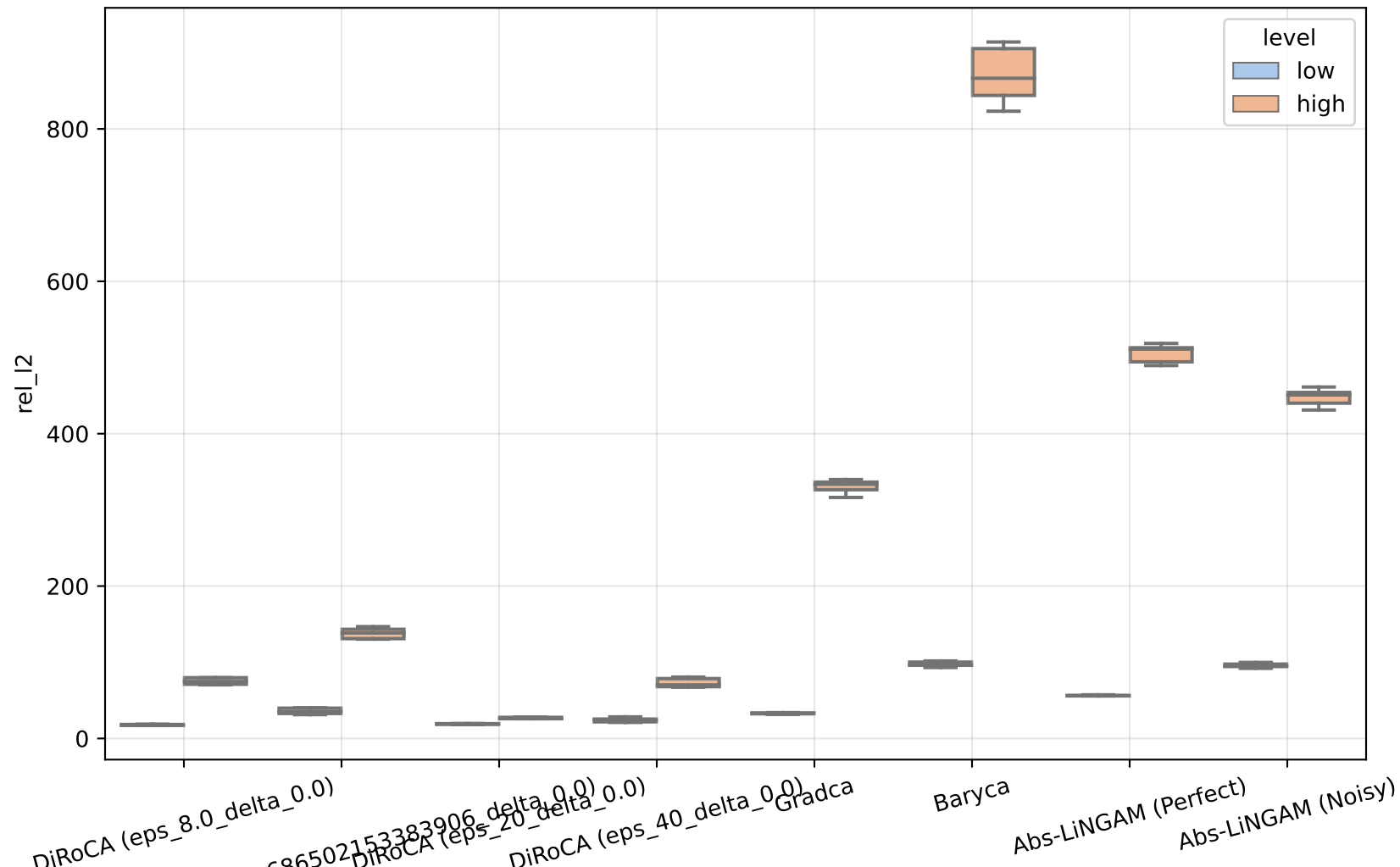
Huber
($\sigma=0.5$)



Relative Error: Rotation



Relative Error: Lighting



Camera Stats: Lighting (High)

DiRoCA	Method	Mean $\pm$ Std	P-Val (vs Best)
	DiRoCA (eps_20_delta_0.0)	26.77 $\pm$ 0.94	(Ref)
	DiRoCA (eps_40_delta_0.0)	72.93 $\pm$ 5.56	0.0000 *
	DiRoCA (eps_8.0_delta_0.0)	75.19 $\pm$ 4.03	0.0000 *
	(eps_61.686502153383906_delta_0.0)	137.99 $\pm$ 6.53	0.0000 *
	Gradca	330.53 $\pm$ 8.36	0.0000 *
	Abs-LiNGAM (Noisy)	447.47 $\pm$ 10.69	0.0000 *
	Abs-LiNGAM (Perfect)	505.20 $\pm$ 11.27	0.0000 *
	Baryca	870.61 $\pm$ 34.88	0.0000 *

Camera Stats: Lighting (Low)

DiRoCA	Method	Mean $\pm$ Std	P-Val (vs Best)
	DiRoCA (eps_8.0_delta_0.0)	17.82 $\pm$ 0.66	(Ref)
	DiRoCA (eps_20_delta_0.0)	18.96 $\pm$ 0.49	0.0000 *
	DiRoCA (eps_40_delta_0.0)	24.13 $\pm$ 2.63	0.0000 *
	Gradca	32.90 $\pm$ 0.83	0.0000 *
	(eps_61.686502153383906_de	35.84 $\pm$ 3.64	0.0000 *
	Abs-LiNGAM (Perfect)	56.09 $\pm$ 0.87	0.0000 *
	Abs-LiNGAM (Noisy)	95.76 $\pm$ 2.68	0.0000 *
	Baryca	97.48 $\pm$ 3.10	0.0000 *

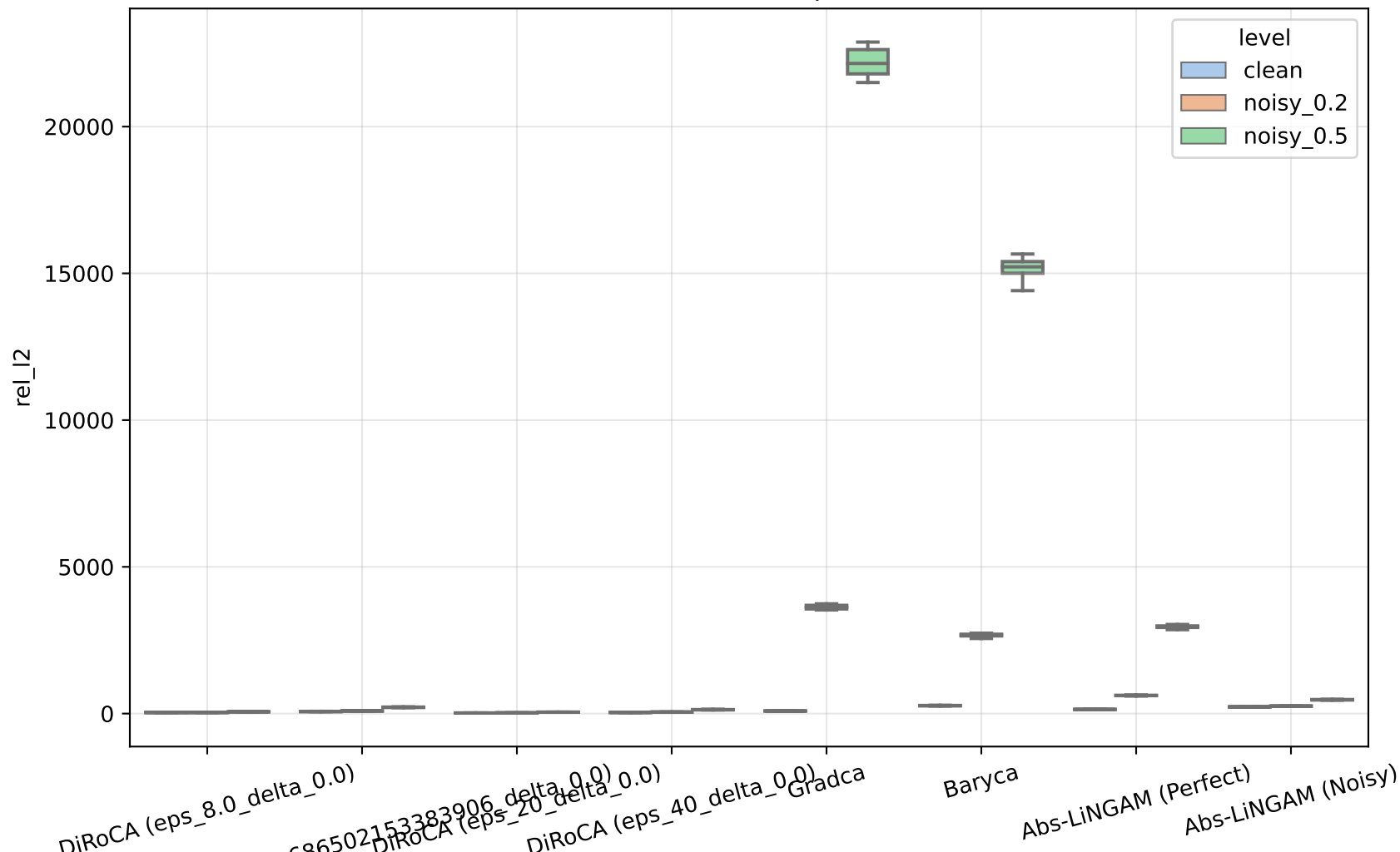
Camera Stats: Rotation (High)

DiRoCA	Method	Mean ± Std	P-Val (vs Best)
	DiRoCA (eps_20_delta_0.0)	30.64 ± 0.72	(Ref)
	DiRoCA (eps_8.0_delta_0.0)	56.71 ± 2.59	0.0000 *
	DiRoCA (eps_40_delta_0.0)	66.60 ± 2.68	0.0000 *
	(eps_61.686502153383906_delta_0.0)	106.40 ± 3.74	0.0000 *
	Abs-LiNGAM (Noisy)	287.80 ± 9.26	0.0000 *
	Abs-LiNGAM (Perfect)	604.37 ± 24.12	0.0000 *
	Gradca	3595.63 ± 653.21	0.0000 *
	Baryca	4131.29 ± 363.07	0.0000 *

Camera Stats: Rotation (Low)

DiRoCA	Method	Mean $\pm$ Std	P-Val (vs Best)
	DiRoCA (eps_20_delta_0.0)	28.02 $\pm$ 0.39	(Ref)
	DiRoCA (eps_8.0_delta_0.0)	52.63 $\pm$ 2.04	0.0000 *
	DiRoCA (eps_40_delta_0.0)	61.48 $\pm$ 4.36	0.0000 *
	(eps_61.686502153383906_de	101.64 $\pm$ 6.20	0.0000 *
	Abs-LiNGAM (Noisy)	276.81 $\pm$ 6.65	0.0000 *
	Abs-LiNGAM (Perfect)	541.15 $\pm$ 13.20	0.0000 *
	Gradca	1776.67 $\pm$ 235.11	0.0000 *
	Baryca	2287.08 $\pm$ 198.31	0.0000 *

Relative Error: Huber (Specific Points)



Huber Stats: Huber (Clean)

DiRoCA	Method	Mean ± Std	P-Val (vs Best)
	DiRoCA (eps_20_delta_0.0)	20.05 ± 0.70	(Ref)
	DiRoCA (eps_8.0_delta_0.0)	35.97 ± 1.70	0.0000 *
	DiRoCA (eps_40_delta_0.0)	38.68 ± 3.20	0.0000 *
	DiRoCA (eps_61.686502153383906_delta_0.0)	69.80 ± 5.04	0.0000 *
	Gradca	87.64 ± 2.17	0.0000 *
	Abs-LiNGAM (Perfect)	141.64 ± 2.44	0.0000 *
	Abs-LiNGAM (Noisy)	227.89 ± 5.78	0.0000 *
	Baryca	270.19 ± 8.69	0.0000 *

Huber Stats: Huber (Noisy_0.2)

DiRoCA	Method	Mean $\pm$ Std	P-Val (vs Best)
	DiRoCA (eps_20_delta_0.0)	24.68 $\pm$ 0.51	(Ref)
	DiRoCA (eps_8.0_delta_0.0)	37.43 $\pm$ 1.51	0.0000 *
	DiRoCA (eps_40_delta_0.0)	53.15 $\pm$ 3.76	0.0000 *
	(eps_61.686502153383906_delta_0.0)	90.63 $\pm$ 5.33	0.0000 *
	Abs-LiNGAM (Noisy)	254.87 $\pm$ 5.71	0.0000 *
	Abs-LiNGAM (Perfect)	614.05 $\pm$ 10.65	0.0000 *
	Baryca	2692.78 $\pm$ 105.27	0.0000 *
	Gradca	3629.19 $\pm$ 242.52	0.0000 *

Huber Stats: Huber (Noisy_0.5)

DiRoCA	Method	Mean $\pm$ Std	P-Val (vs Best)
	DiRoCA (eps_20_delta_0.0)	48.17 $\pm$ 0.64	(Ref)
	DiRoCA (eps_8.0_delta_0.0)	60.33 $\pm$ 1.73	0.0000 *
	DiRoCA (eps_40_delta_0.0)	132.86 $\pm$ 7.48	0.0000 *
	DiRoCA (eps_61.686502153383906_delta_0.0)	214.91 $\pm$ 8.42	0.0000 *
	Abs-LiNGAM (Noisy)	470.84 $\pm$ 8.45	0.0000 *
	Abs-LiNGAM (Perfect)	2951.69 $\pm$ 51.59	0.0000 *
	Baryca	15338.79 $\pm$ 628.99	0.0000 *
	Gradca	22207.40 $\pm$ 1501.67	0.0000 *

Robustness Curve: Error vs Noise Intensity (Huber $\alpha=1.0$)

